

Telematics Driver Behaviour Scoring Evaluation

Expert Review Questionnaire by Matthew Bowyer

Thank you for taking the time to participate in this expert review.

The purpose of this review is to assess whether the scoring telematics output is credible, operationally useful, and aligned with real-world fleet management expectations.

All outputs shown to you are anonymised. No driver names, client names, or commercially sensitive identifiers are included.

The intention is to use expert feedback to either refine and improve the scoring framework or validate that the system meets professional fleet management standards.

** Indicates required question*

1. What is your current Role? *

2. Years in fleet/telematics industry? *

Mark only one oval.

☐ 0-2

☐ 3-5

☐ 6-10

☐ 11+

3. What fleet type do you work with? *

Mark only one oval.

- ☐ Passenger
- ☐ Light commercial
- ☐ Heavy commercial
- ☐ Mixed/Multiple fleet types

4. What are your approximate vehicles managed? *

Mark only one oval.

- ☐ 1-50
- ☐ 51-200
- ☐ 201-1000
- ☐ 1000+

5. Do you currently use a driver score/rating system? *

Mark only one oval.

- ☐ Yes
- ☐ No

Materials - You have been provided with three outputs to review.

A. Behaviour Category Weight Structure

The final driver behaviour score is built from six ranked measures derived from aggregated telematics data.

1. Total driving exposure (drive hours): 5%
2. Night driving ratio: 10%
3. Harsh manoeuvres (cornering, braking, acceleration): 15%
4. Speeding events: 15%
5. Power and compliance violations: 30%
6. AI-based attention and situational risk indicators: 25%

Step 1: Relative Scoring

Each vehicle is compared to all other vehicles in the population.

For every metric:

- Performance is converted into a value between 0 and 1
- The score reflects where the vehicle sits relative to the group

This means the system is population-based, not threshold-based.

Step 2: Weighted Combination

The six relative scores are multiplied by their assigned weights and combined into a single UBI proxy score.

This produces one overall risk score per vehicle.

Risk Classification

Vehicles are classified into (based on distribution):

- Low Risk → Bottom third
- Medium Risk → Middle third
- High Risk → Top third

This ensures risk levels reflect relative performance within the fleet rather than arbitrary fixed cutoffs.

B. Event-to-Category Mapping Table

This table explains:

- What each event represents operationally.
- Whether the event comes from the telematics device or AI camera.
- Which behavioural category the event contributes to.

This ensures transparency in how raw events influence the final score.

Event Name	Short Description	Source Device	Affected rank column
Speeding	Vehicle speed exceeds client-set max (default 130 km/h)	Telematics	Speeding
Harsh Braking	Sudden decrease in speed	Telematics	Harsh
Harsh Acceleration	Sudden increase in speed	Telematics	Harsh
Harsh Cornering	Sharp direction change	Telematics	Harsh
Night Driving Ratio	% driving between 22:00–05:00	Telematics	Night Ratio
Seatbelt Off	Seatbelt not detected (ADC or CAN)	Telematics	Violations
Power Disconnect	External battery disconnected	Telematics	Violations
Camera Covered	Camera lens obstruction detected	AI Camera	Violations
Lane Departure	Unsafe lane deviation detected	AI Camera	Attention
Forward Collision Risk	Unsafe closing distance to vehicle ahead	AI Camera	Attention
Headway Monitoring	Unsafe following distance	AI Camera	Attention
Phone Usage	Phone use while driving	AI Camera	Attention
Driver Distraction	Visual distraction detected	AI Camera	Attention
Eye Closure	Prolonged eye closure detected	AI Camera	Attention
Fatigue	Fatigue pattern detected	AI Camera	Attention
Yawning	Yawning behaviour detected	AI Camera	Attention
Smoking	Smoking detected	AI Camera	Attention
Pedestrian Collision Risk	Pedestrian impact risk detected	AI Camera	Attention

C. Risk Ranking Tables

These include:

- Top 10 highest-risk vehicles for the month
- Top 10 lowest-risk vehicles
- A breakdown of behavioural category contributions for each vehicle

These rankings simulate how a fleet manager would:

- Prioritise coaching
- Investigate incidents
- Review driver performance

Top 10 High-Risk Vehicles

vehicle_id	drive_hours	night_ratio	harsh_total	speed_total	power_events	attention_total	ubi_proxy_score	vehicle_behaviour_class
365598883	179.511111	0.107644	139	78	81	10833	0.838276	High
338306241	202.011944	0.097927	18	28	259	18612	0.837586	High
365599055	186.381111	0.101793	29	9	121	13215	0.820259	High
454260123	146.435278	0.102366	32	24	129	11318	0.817759	High
507302529	159.878056	0.147248	18	17	176	10319	0.808793	High
307989723	133.258333	0.173131	26	2	4	9287	0.798621	High
334148753	230.435000	0.103583	116	73	2	5544	0.790517	High
333545827	483.996389	0.132929	133	2	58	198	0.784655	High
371660225	109.795833	0.018054	29	18	11	15630	0.783879	High
336854753	242.640000	0.008467	120	40	247	10563	0.779655	High

Top 10 Low-Risk Vehicles

vehicle_id	drive_hours	night_ratio	harsh_total	speed_total	power_events	attention_total	ubi_proxy_score	vehicle_behaviour_class
376677144	0.000000	0.000000	0	0	0	0	0.236379	Low
378317132	0.000000	0.000000	0	0	0	0	0.236379	Low
509636643	0.553056	0.000000	0	0	0	0	0.238017	Low
509500466	4.272500	0.000000	0	0	0	0	0.241293	Low
443664708	8.836389	0.000000	0	0	0	0	0.242155	Low
459505629	29.329444	0.000000	0	0	0	0	0.245431	Low
459503546	77.516389	0.000000	0	0	0	0	0.261983	Low
509387327	2.853611	0.000000	2	0	0	0	0.267328	Low
509637190	0.490278	0.000000	5	0	0	0	0.273793	Low
457001649	0.671111	0.031457	0	0	0	0	0.275948	Low

6. All questions below are based on the above information and examples. *

Do the top 10 and bottom 10 rankings look credible?

1 (Not credible) - 5 (Credible)

1	2	3	4	5
				

7. What (if anything) looked clearly wrong or suspicious in the ranking? *

8. Are any of the six behaviour categories inappropriate, overemphasised, or missing? *

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9. Do the current weights for the six behaviour categories reflect real-world risk importance? *

1 (Not at all) - 5 (Perfectly)

1	2	3	4	5
				

10. If you could change the weights, what would you adjust and why? *

11. Would you trust this score to prioritise coaching, escalate investigations, or inform incentives/disciplinary decisions? *

Final Remarks and Appreciation

Thank you for taking the time to complete this evaluation.

Your feedback provides essential real-world insight into whether this driver scoring framework reflects operational fleet risk and management priorities. Expert input such as yours strengthens the practical relevance and defensibility of the research.

12. If you have any additional comments or reflections not covered in the questions above, please include them here:

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